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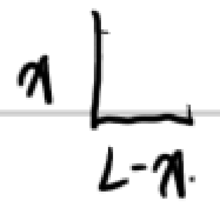
P1. (a) $F = \frac{x}{L} mg$



$W = \int_0^l F\, dx = \frac{mg}{L}\int_0^l x\, dx = \frac{1}{2} mgl$

(b) $m(x) = \int_0^x \lambda(x)\, dx = \int_0^x \lambda_0 x(l-x)\, dx = -\frac{1}{3}\lambda_0 x^3 + \frac{1}{2}\lambda_0 x^2$ $\quad m(l) = m \Rightarrow l^3 = \frac{6m}{\lambda_0}$

$F = m(x) g = -\frac{1}{3}\lambda_0 x^3 g + \frac{1}{2}\lambda_0 x^2 g$

$W = \int_0^l F\, dx = \lambda_0 g \int_0^l \left(-\frac{1}{3}x^3 + \frac{1}{2}x^2\right) dx$

$= \lambda_0 g \left(-\frac{1}{12} l^4 + \frac{1}{6} l^4\right)$

$= \frac{1}{2} mgl$

P2. $\frac{1}{3}\rho g\left(\pi R^2 \cdot H \cdot \frac{1}{3}\right) = mg \Rightarrow \frac{1}{9}\rho\, \pi R^2 H = m$. take the ground as FoR

$\frac{1}{9}\pi R^2 H = \frac{1}{3}\pi\left(R\frac{h}{H}\right)^2 h \Rightarrow h = \frac{H}{\sqrt[3]{3}}$

$F_f = \rho g \cdot \frac{1}{3}\pi\left(R\frac{h-x}{H}\right)^2 (h-x)$

$= \frac{1}{3}\rho g \pi R^2 \frac{1}{H^2}(h-x)^3$

$W_m + \int_0^h F_f \cdot dx = mgh$

$W_m = mgh - \int_0^h F_f\, dx = mgh - \frac{1}{3}\rho g\pi \frac{R^2}{H^2}\int_0^h (h-x)^3 dx$

$= mgh - \frac{1}{3}\rho g \pi \frac{R^2}{H^2}\cdot\frac{1}{4}h^4$

$= \frac{1}{12\cdot\sqrt[3]{3}}\rho g \pi R^2 H^2$

take the ground as FoR.

P3 (a) $mgx = \frac{1}{2}k x_m^2$ $\quad x_m = \frac{2mg}{k} = \frac{g}{300}$ m $\quad x_m = 0.033$ m.

$F_m = k x_m = 98$ N. direction: upwards

(b) $mgx = \int_0^x F\, dx \Rightarrow mgx = \int_0^x 3(x + 160x^3)\, dx = \left(\frac{3}{2}x^2 + 120x^4\right)\times 10^3$

$300 x_m + 24000 x_m^3 = g$. $\quad x_m = 0.03$ m

$F_m = 3\times10^3 (x_m + 160 x_m^3) = 104.83$ N.

P4. $\begin{cases} mg\sin\alpha \le Ax\, mg\cos\alpha & ① \\ \frac{1}{2}mv_0^2 = mgx\sin\alpha + \int_0^x \mu mg\cos\alpha\, dx & ② \end{cases}$

$\int_0^x \mu mg\cos\alpha\, dx = mg\cos\alpha \int_0^x Ax\, dx = mg\cos\alpha \cdot \frac{1}{2}Ax^2 = \frac{1}{2}A mg\cos\alpha\, x^2$

$\frac{1}{2}mv_0^2 = mgx\sin\alpha + \frac{1}{2}A mg\cos\alpha\, x^2$

for ①: $x \geq \frac{\tan\alpha}{A}$

so $mgx\sin\alpha + \frac{1}{2}Amg\cos\alpha\, x^2 = \frac{1}{2}mv_0^2 \geq \frac{3}{2}mg\frac{\sin^2\alpha}{A\cos\alpha}$

$v_0^2 \geq \frac{3g\sin^2\alpha}{A\cos\alpha}$

P5. $\frac{1}{2}mv_0^2 = \int_0^{\frac{1}{2}L_0} f\,dx$. $\quad f = u\frac{x}{L_0}mg$

so $\int_0^{\frac{1}{2}L_0} f\,dx = \int_0^{\frac{1}{2}L_0} \frac{u}{L_0}mg\,x\,dx = \frac{u}{L_0}mg\int_0^{\frac{1}{2}L_0} x\,dx = \frac{1}{8}umgL_0$

$\frac{1}{2}mv_0^2 = \frac{1}{8}umgL_0 \Rightarrow u = \frac{4v_0^2}{gL_0} \quad v_0 = \sqrt{\frac{ugL_0}{4}} = \frac{1}{2}\sqrt{ugL_0}$

P6.

Aa. $\bar{r} = (x, 0, 0)$

$W_1 = \int F_1(r)\cdot dr = \int_{-1}^{1} x^2 z\,dx = 0$

Ab. $\bar{r} = (x, x^2-1, 0)$ $\quad d\bar{r} = dx\cdot(1, 2x, 0)$

$\bar{F}_1 = (0, -x(x^2-1), 5)$

$W_2 = \int \bar{F}_1(r)\cdot dr = \int_{-1}^{1} -x(x^2-1)\,2x\,dx = \frac{8}{15}$

Ba. $r = (x, 0, 0)$

$\bar{F}_2 = (-2x, 0, 0)$

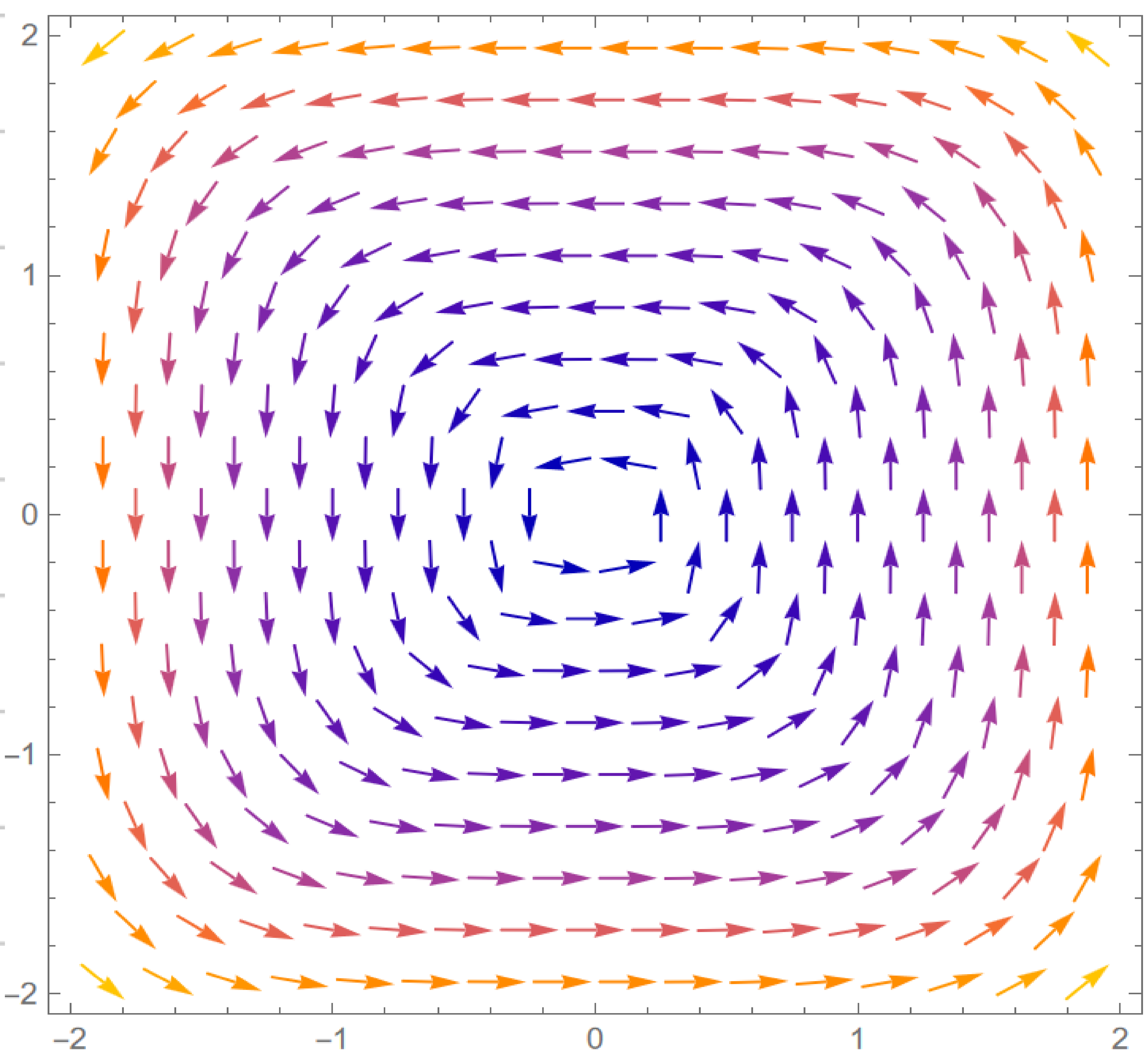
$W_3 = \int \bar{F}_2\,d\bar{r} = \int_{-1}^{1} -2x\,dx = 0$

Bb. $\bar{r} = (x, x^2-1, 0)$

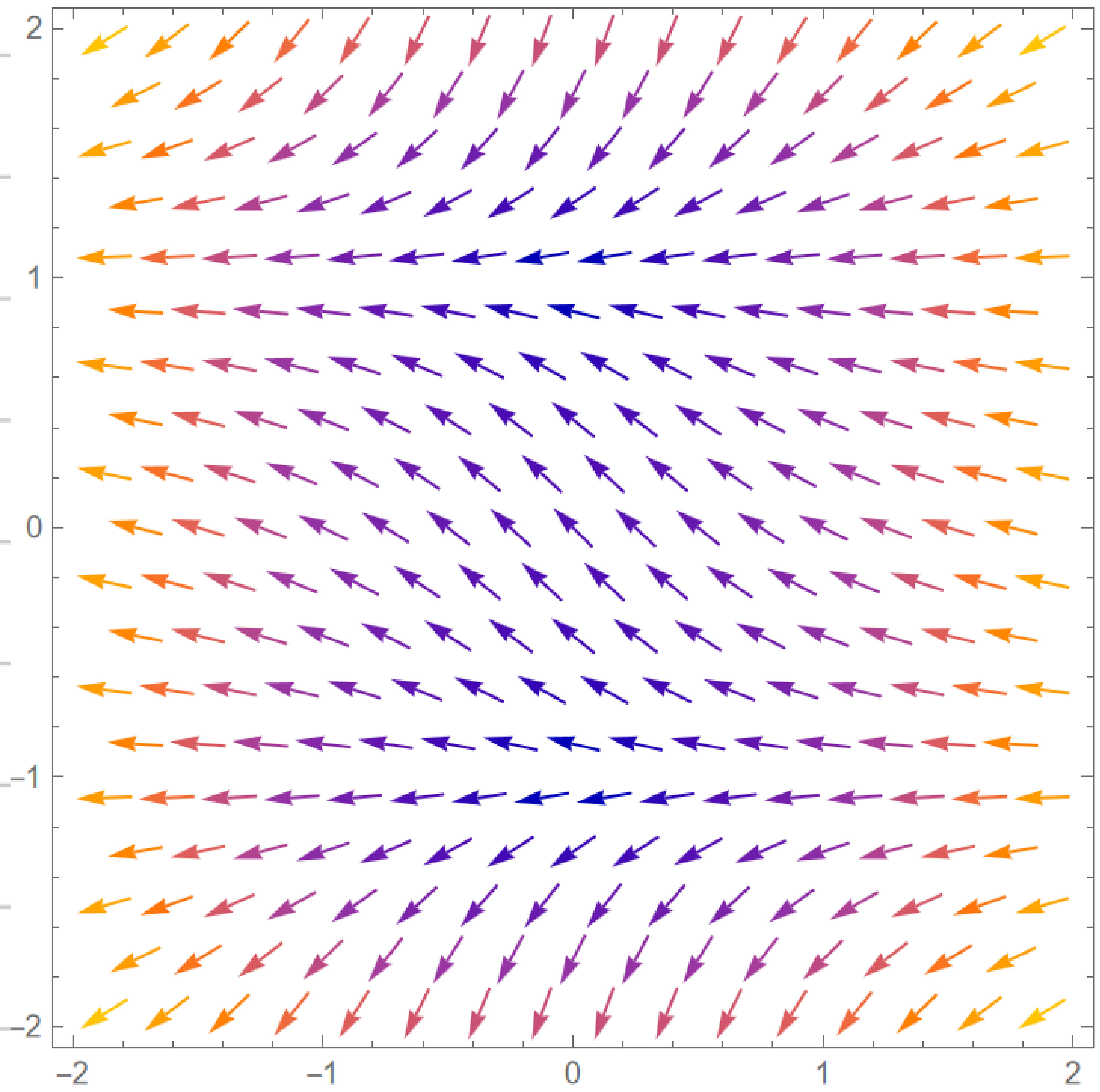
$\bar{F}_2 = (-2x, 0, x^2-1-x(x^2-1)) = (-2x, 0, -x^3+x^2+x-1)$

$W_4 = \int \bar{F}_2\,d\bar{r} = \int_{-1}^{1} -2x\,dx = 0$

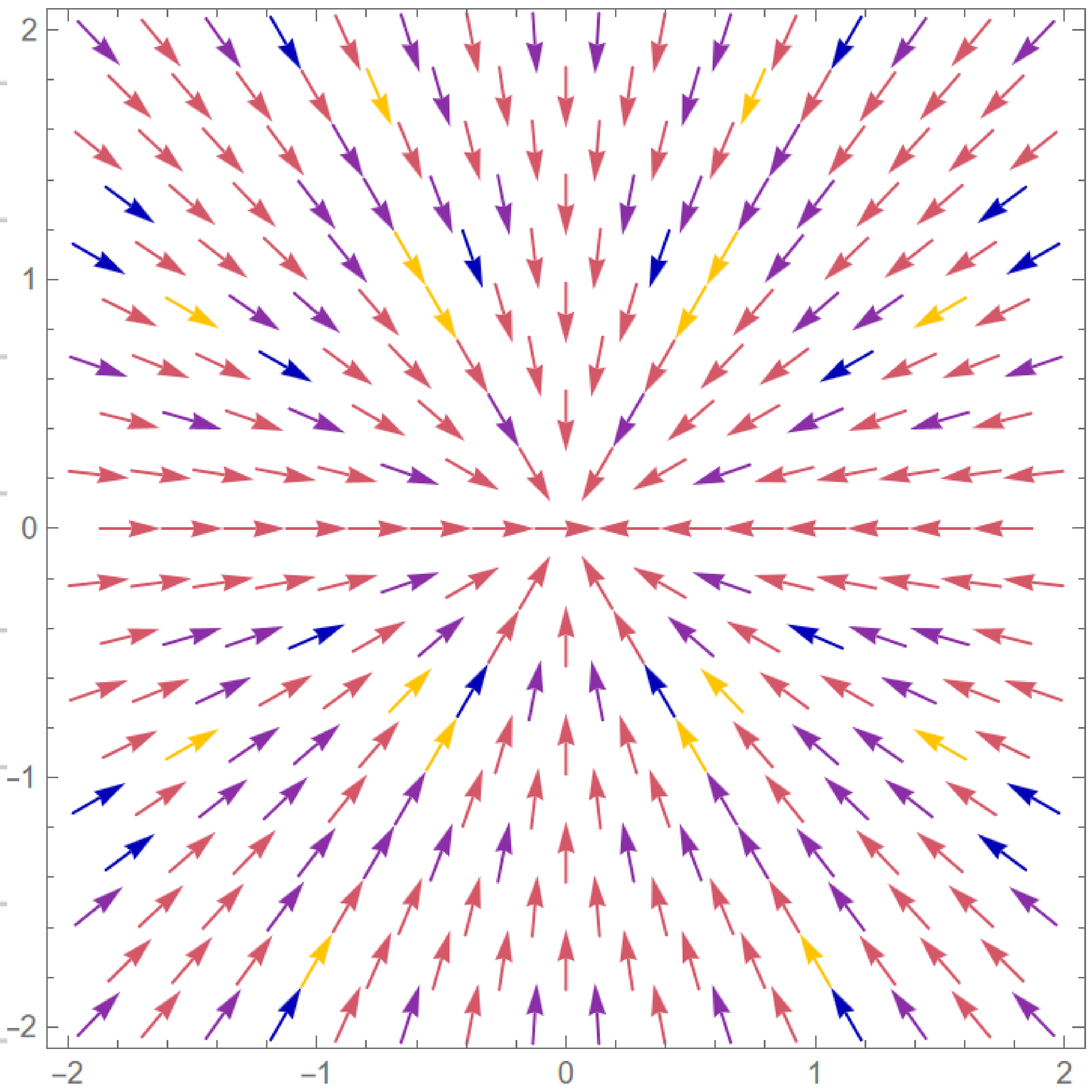
P7. (a)



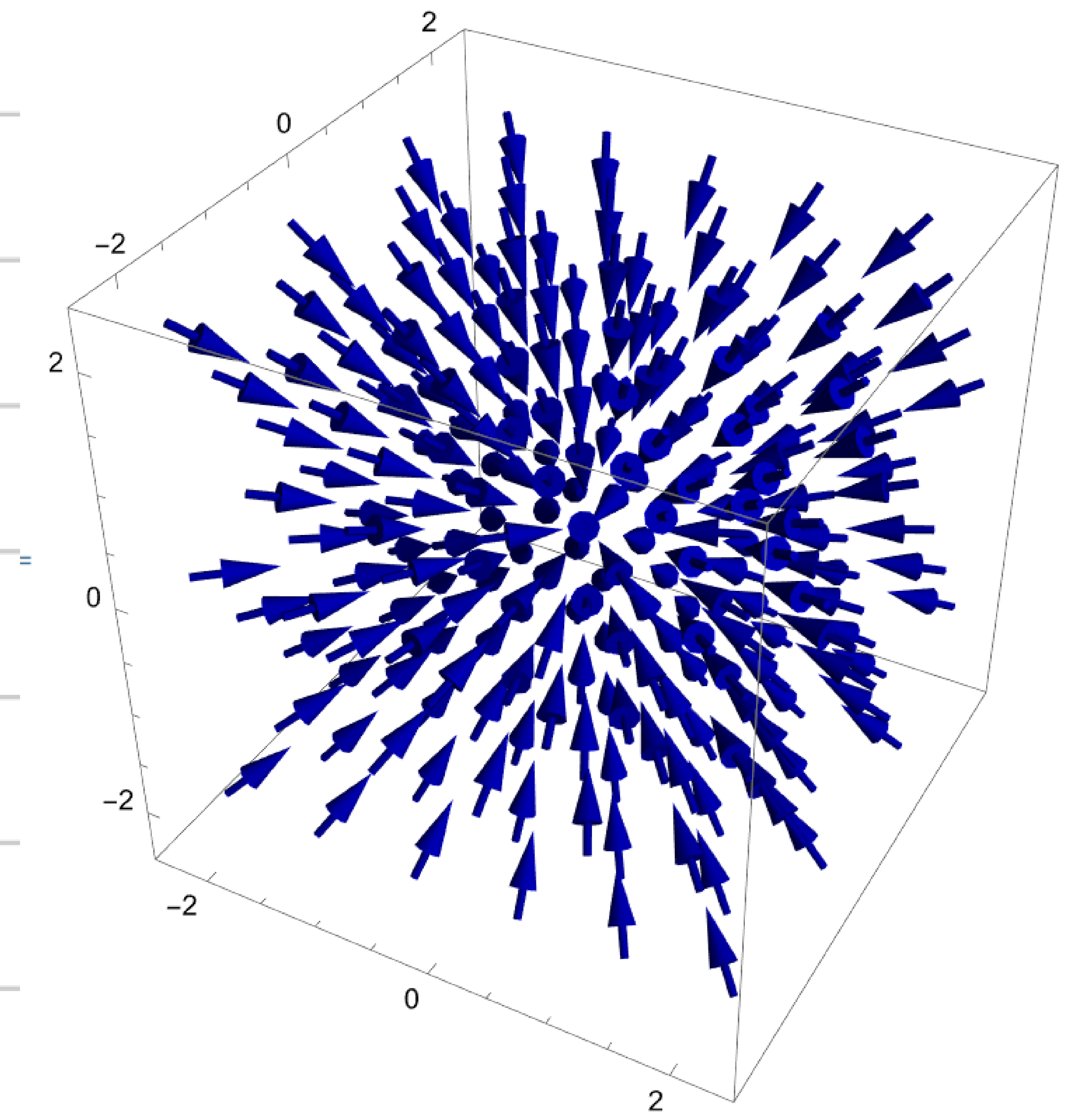
(b)



(c) 2D



3D



(d)